

## TME 9 - Recommender Systems

### Recommendation and the Surprise Module

The module surprise is a module specifically dedicated to recommendation systems, following the conventions of `scikit-learn`.

The objective of this TME is to explore and compare different basic recommendation algorithms using this module. You will use the MovieLens dataset (the small version, the cleaned dataset restricted to movies/users with more than 10 ratings, is provided as a complement to this assignment).

To load the data, use `surprise.Dataset.load_from_file("movielenssmall.csv", surprise.Reader())`. The other file contains movie titles.

First, conduct a preliminary analysis of the data by plotting histograms of ratings, the number of reviews per movie/user, and the average ratings of movies/users. Also, visualize the pairwise distance matrix between users using Euclidean and cosine distances. Compute the sparsity of the dataset.

Next, split your dataset into training/testing sets (by randomly removing a percentage of ratings from the rating matrix). Test and compare at least the following algorithms (by reading the documentation to understand the differences and testing with various parameter settings) :

- `BaselineOnly`
- `KNNBasic`
- `KNNWithMeans`
- `SVD`
- `NMF`

For matrix factorization-based methods, you can also plot the 2D latent representations of movies using a dimensionality reduction algorithm such as t-SNE (e.g., via `tensorboard`).

It may also be interesting to cluster the latent representations of movies and interpret the resulting clusters.